GlucoTwin AI

A Time-Series Digital Twin Proof of Concept

Problem statement

Demonstrate a virtual patient model that combines historical context and incoming time-series readings to forecast a future glucose value.

Healthcare use case

A simulated patient has glucose, activity, meal estimate, sleep and heart-rate readings. The model forecasts glucose 30 minutes ahead and displays trends.

Technical stack

Python, FastAPI, pandas, NumPy, scikit-learn, joblib and optional React charts.

AI/ML model

Random Forest regression baseline with lag and rolling features. A chronological train/test split is used; MAE and RMSE are reported.

Prototype endpoints

GET /health; GET /twin/demo-patient; GET /forecast/demo-patient?horizon_minutes=30.

Evaluation

Run python ml/train.py and insert the actual MAE/RMSE from ml/metrics.json in the final presentation. Never claim unmeasured performance.

Limitations and ethics

Synthetic data only; no clinical validation. Outputs are not medical advice. Do not use for diagnosis or treatment.

Submission checklist

Public GitHub repo, complete README, architecture PDF, presentation PDF/PPTX, MIT licence and unlisted 15–20 minute demo video.